Evaluating the Determinants of Employee Compensation at HCL Technologies

An econometric analysis of salary packages using employee-level data

DOM207 / DOM3007 / DOM6401 — Mini-Project 2

Prepared for: HCL Technologies

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1. Introduction

HCL Technologies has engaged DOM207-DOM3007-DOM6401 Inc. to carry out an independent evaluation of the salary packages paid across its workforce. The firm has provided a confidential extract of 526 employee records covering human-capital endowments (education, professional experience and firm tenure), household and demographic characteristics (gender, marital status, dependents, parental education, siblings), geographic location, and the employee's job function and division within the IT organisation. The extract also carries the average hourly salary of each employee, which is the outcome the client wishes to understand.

The purpose of this engagement is therefore explanatory rather than purely descriptive. The client wants to know (i) which characteristics are systematically associated with pay, (ii) how large those associations are, and (iii) whether the pay structure contains features - gender differentials, marriage premiums, geographic or functional premia - that may require managerial attention. Our objective is to estimate a defensible econometric model of compensation, to test the mechanisms suggested by economic theory, and to translate the estimated coefficients into magnitudes that a non-technical audience can act on.

The human-capital earnings function of Mincer (1974) provides the organising framework. Employees are assumed to be paid according to the productive characteristics they bring to the firm - schooling, general experience and firm-specific experience - with additional allowances for household circumstances, location and the internal labour market in which the employee works. We begin from that baseline, then deliberately extend it in the two directions the client brief flags as critical: non-linear (quadratic) effects that capture diminishing returns, and interaction effects that allow the returns to one characteristic to depend on another. Supplementary analysis examines model specification, multicollinearity and heteroskedasticity, and several alternative specifications are estimated to gauge the subsidiary influences of the study variables.

The remainder of the report is organised as follows. Section 2 documents the data-processing decisions. Section 3 presents the visual evidence. Section 4 states the hypotheses and their theoretical justification. Section 5 develops and diagnoses the models and reports the results. Section 6 draws the recommendations, and Section 7 lists the references.

2. Data Processing

The raw file ITEmp.xlsx contains one sheet of 526 employee records and 24 analytic variables, together with an employee identifier. The data dictionary distinguishes eight continuous or count variables (sal, edu, exper, ten, dep, momed, daded, sib) from sixteen binary indicators grouped into three families: demographic and household, geographic, and IT job function/division. No new observations were created and no value was altered on the basis of an assumption about what the 'correct' figure should have been; every value we could not reconcile with the data dictionary was treated as missing.

Three classes of problem were identified in the raw extract.

- Non-numeric entries in numeric fields. Eight cells contained text where a number was required (for example 'SS0' in edu, 'aaa' in south, 'acb' in sal and 'ab' in momed). These are transcription errors and were set to missing.
- Out-of-range values. Twenty-one cells contradicted the data dictionary. Binary indicators can only take the values 0 or 1, yet the file contained values such as 100 and 1111; education cannot be 1,001 years, nor salary 1,111 USD per hour, nor an employee have 333 siblings. Each of these was set to missing.
- Missing values. Twenty-two cells were already empty in the raw file.

Because invalid and missing cells together account for only 51 of the 12,624 analytic cells, listwise deletion would have discarded far more information than it protected. We therefore imputed: continuous variables with the column median and binary variables with the column mode. Imputation affected 51 cells, or 0.40% of the analytic matrix. To demonstrate that this choice is not driving the results, a complete-case dataset (482 employees) is carried through the analysis as model M10; the conclusions are unchanged.

Four variables were derived for the modelling stage: the natural logarithm of salary and the squares of experience, tenure and education, which the quadratic specifications require. The cleaned analysis file is ITEmp_Cleaned.csv and the audit trail is reproduced in Table 1.

Table 1. Data-cleaning audit by variable (n = 526 raw records).

Variable	Role	Missing in raw	Non-numeric	Out of range	Total set to missing
sal	dependent	2	1	4	7
edu	continuous	1	1	5	7
exper	continuous	2	0	0	2
ten	continuous	0	0	0	0
dep	continuous	0	0	0	0
momed	continuous	3	1	1	5
daded	continuous	2	0	1	3
sib	continuous	1	0	4	5
nonprev	binary	0	0	0	0
female	binary	0	0	0	0
married	binary	0	0	0	0
metro	binary	1	1	0	2
norcen	binary	1	0	1	2
south	binary	5	1	2	8
west	binary	0	2	1	3
itreq	binary	4	0	1	5
itprog	binary	0	0	1	1
itqual	binary	0	1	0	1
itarch	binary	0	0	0	0
itai	binary	0	0	0	0
itclient	binary	0	0	0	0
itrd	binary	0	0	0	0
ittech	binary	0	0	0	0
wrkhm	binary	0	0	0	0

3. Data Visualization

The figures below are restricted to the relationships that actually inform the modelling choices. Figure 1 motivates the choice of functional form; Figure 2 anticipates the collinearity structure; Figures 3 and 4 display the functional and geographic differentials; Figure 5 examines the human-capital variables; Figure 6 the group contrasts; and Figure 7 the experience-tenure relationship.

Figure 1. The dependent variable is right-skewed; the log transform restores near-symmetry

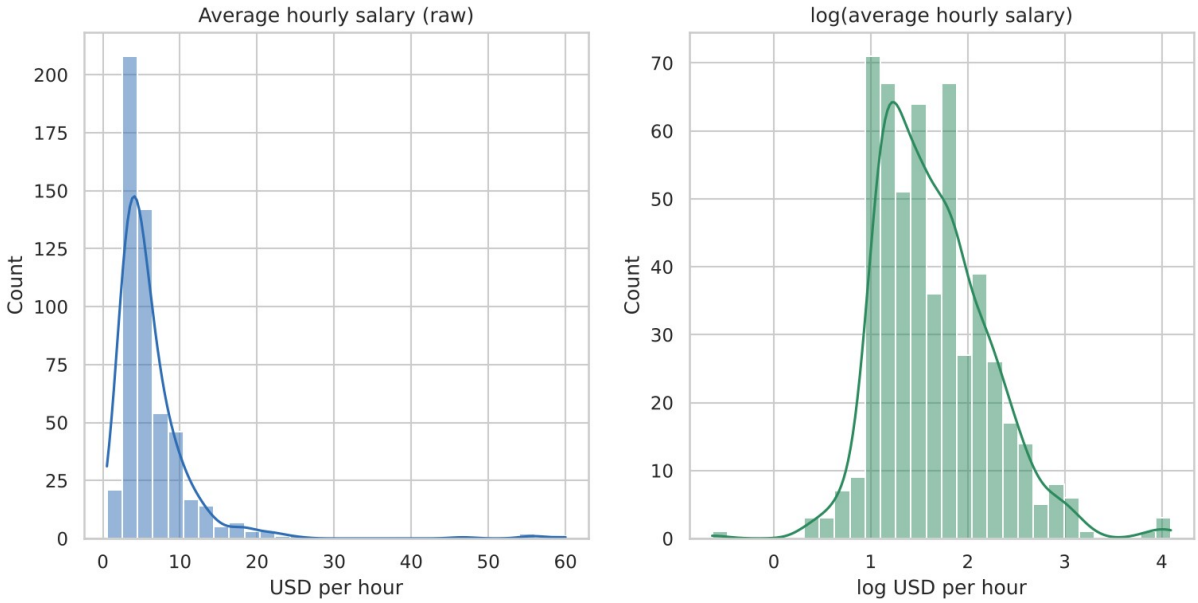


Figure 1. Average hourly salary in levels (left) and in logarithms (right). The level distribution is strongly right-skewed (skewness 5.45); the logarithm is far closer to symmetric (0.77), which is why the models below are estimated on log salary.

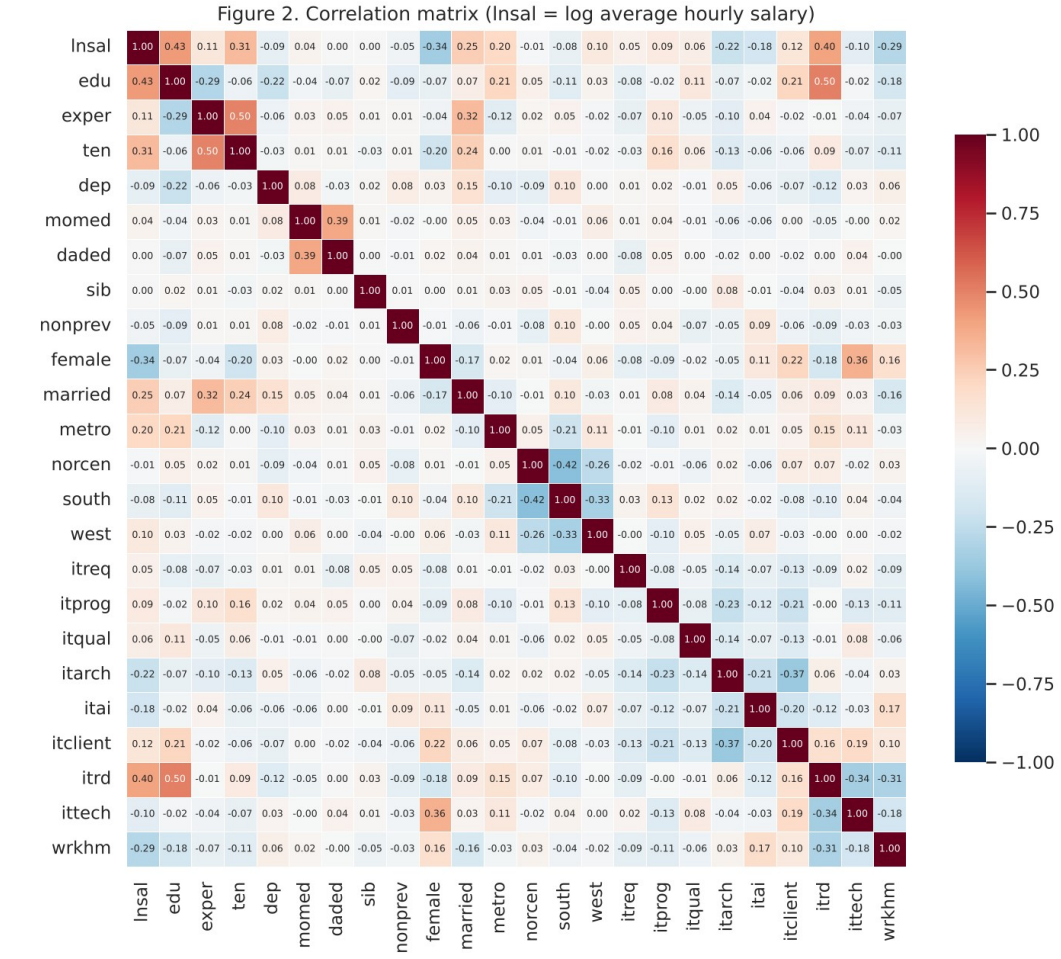


Figure 2. Correlation matrix. Salary is most strongly correlated with education (0.43), the R&D division (0.40), gender (-0.34) and tenure (0.31). The notable feature for modelling is the 0.50 correlation between education and working in R&D, and the 0.50 correlation between experience and tenure: these are the relationships that could generate collinearity.

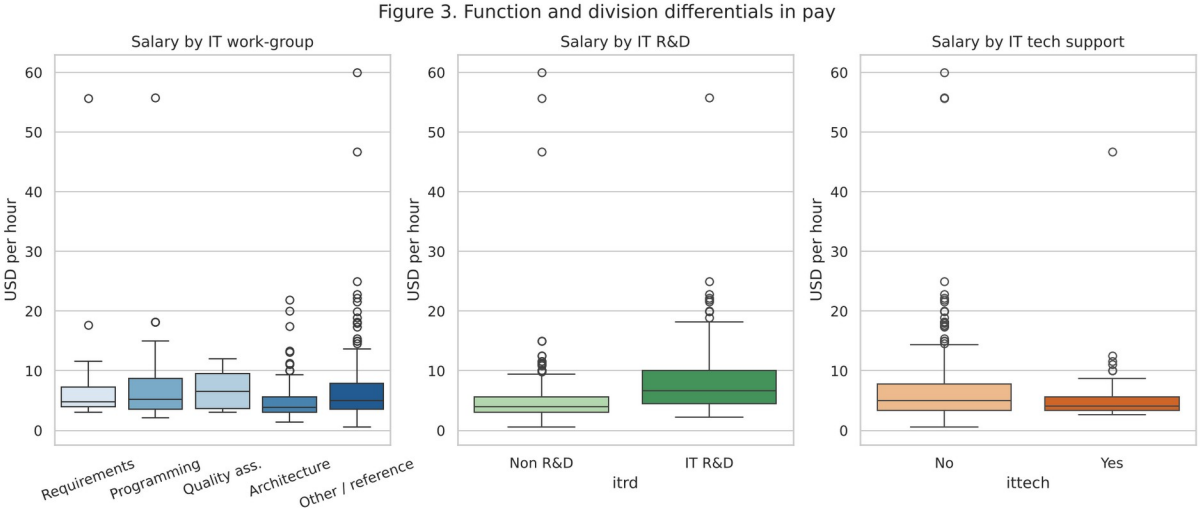


Figure 3. Pay by IT work-group and division. R&D employees earn visibly more; employees in IT architecture and AI-exposed roles earn less, and the differences are large enough to be worth formal testing.

Figure 4. Geographic pay differentials

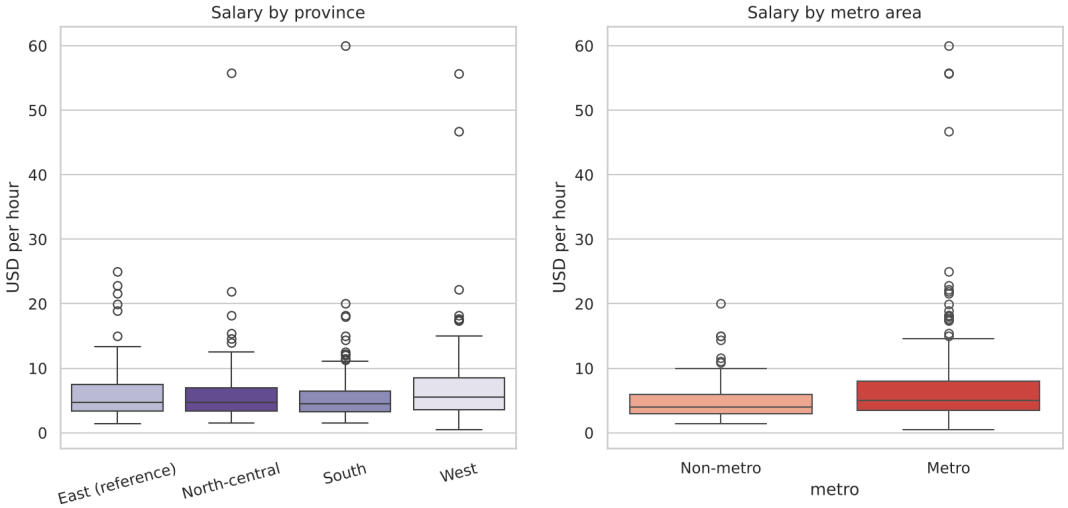


Figure 4. Pay by province and by metro location. Metro employees earn a clear premium. The three province indicators (with the eastern province as the reference category) show much smaller and less orderly differences.

Figure 5. Salary vs human-capital stock
solid = linear fit; dashed = quadratic fit where shown

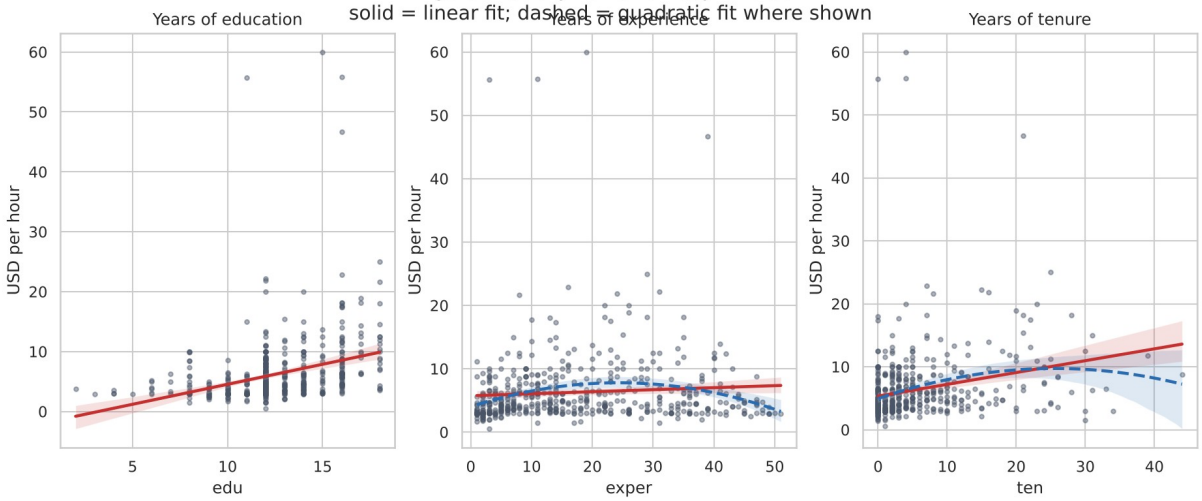


Figure 5. Salary against the human-capital variables. Education and tenure show upward sloping profiles; the experience profile flattens at higher levels, which is the visual counterpart of the concave quadratic estimated in Section 5.3.

Figure 6. Salary distributions across demographic and work-mode groups

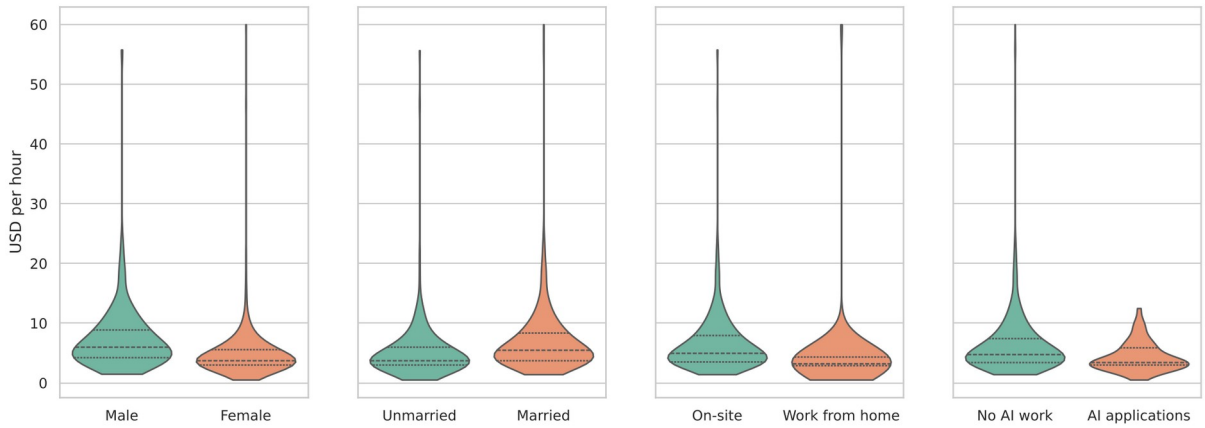


Figure 6. Salary distributions by gender, marital status, work mode and AI exposure. Men, married employees and on-site employees sit higher in the distribution; AI-exposed roles sit lower.

Figure 7. The experience-tenure collinearity and the non-linear experience-salary profile



Figure 7. Experience and tenure are strongly related (left) - an employee cannot have more years at the firm than years of total experience - while average salary rises with experience up to roughly 25-30 years and then levels off or declines (right).

Descriptive statistics for every variable, and mean salaries by group, are reported in the accompanying output tables (output_tables/descriptive_stats.csv and group_means.csv).

4. Hypotheses

The theoretical justification for each regressor, and the direction we expect, follows from standard labour economics. Education and experience are the classical arguments of the human-capital earnings function (Becker 1964; Mincer 1974): schooling raises productivity and

therefore pay, and labour-market experience does too, but at a diminishing rate because returns to additional experience fall as the stock of human capital grows. Tenure is the firm-specific component of human capital and should attract a premium because it is not fully transferable to other employers. Household variables enter as proxies for commitment and mobility: married employees may earn a premium (the widely documented 'marriage premium'), and dependents may raise or lower pay depending on whether they proxy responsibility or reduced mobility.

Gender enters because of the possibility of wage discrimination on the part of employers or of supply-side differences that our controls do not fully capture. Location enters through compensating differentials: a metropolitan posting should command a premium to offset a higher cost of living. The IT function and division indicators capture internal labour markets with different scarcities of skill, and the parental-education and sibling variables are included as family-background controls that could otherwise contaminate the education coefficient. Table 2 states each hypothesis, its expected sign and its theoretical basis.

Table 2. Hypotheses, expected signs and theoretical justification.

H#	Hypothesis	Expected sign	Theoretical justification
H1	Education raises pay	+	Human-capital theory: schooling raises productivity.
H2	Experience raises pay but with diminishing returns	+	Mincer: general experience raises pay, with diminishing returns.
H2b	The experience-pay profile is concave	-	Concavity of the experience-earnings profile.
H3	Firm-specific tenure raises pay	+	Firm-specific human capital is not perfectly portable.
H4	Women earn less than comparable men	-	Statistical discrimination / differential supply behaviour.
H5	Married employees earn a premium	+	Marriage premium (specialisation and commitment).
H6	Metro employees earn a premium	+	Compensating differential for metropolitan cost of living.
H7	Pay differs across IT work-groups and divisions	+	Internal labour markets differ in scarcity of skills.
H7b	R&D and tech-support divisions shift pay	+	Divisional premia for R&D and tech-support skills.
H8	Non-privileged employees earn less	-	Non-privileged status proxies weaker outside options.
H9	Working from home is associated with lower pay	-	Remote work may carry a pay penalty.
H9b	AI-exposed IT work is associated with lower pay	-	AI-exposed work signals obsolescence risk.
H10	Parental education and siblings matter	+	Family background proxies omitted ability.
H11	The education premium differs by gender	-	Returns to schooling may differ by gender.
H12	Education and experience are complements	+	Education and experience are complements in production.
H13	The marriage premium differs by gender	-	The marriage premium may differ by gender.

5. Analysis

5.1 Model specification

The dependent variable is the natural logarithm of average hourly salary. Figure 1 shows why: salary in levels spans 0.53 to 60.00 USD per hour with a skewness of 5.45, whereas the logarithm is close to symmetric. Using the logarithm also makes the coefficients directly interpretable as proportional effects - a coefficient of b on a regressor means that a one-unit increase in that regressor is associated with approximately 100 x b percent higher salary. Estimating the same model in levels (model M9) raises the question of whether the log transformation is imposed rather than earned; it is not, because the level model fits far worse (adjusted R-squared 0.248 against 0.495).

The model is estimated by ordinary least squares, which is appropriate here because salary is a continuous outcome and the objective is to characterise the conditional mean. OLS requires that the relationship be linear in the parameters, that the regressors be non-collinear, that the errors have constant variance, and, for the classical standard errors to be valid, that the errors be normal. Every one of these conditions is tested explicitly in Section 5.6 and, where it fails, remedied or reported.

Because the quadratic and interaction terms involve products of regressors, the continuous variables entering such terms are mean-centred before the products are formed. This is not cosmetic: without centring, education alone produces a variance-inflation factor above 80, which is purely an artefact of collinearity between x and x-squared. Centring removes that artificial collinearity while leaving the fitted values, R-squared and every substantive conclusion unchanged, and it has the useful side effect that the linear coefficient on, say, experience is read as the effect at the average level of experience.

5.2 Main results

Table 3 reports the nested sequence of models. M1 contains only the human-capital variables; M2 to M5 progressively add demographic and household controls, geographic controls, the IT function and division controls, and the family-background variables; M6 adds the quadratic terms and M7 the interactions. The preferred specification is M7.

*Table 3. Determinants of log average hourly salary (n = 526). Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.10.*

variable	M1 Human capital	M2 + Demographics	M3 + Geography	M4 + Job function	M5 Full main effects	M6 + Quadratics	M7 + Interactions
Intercept	1.6434***	1.7165***	1.5782***	1.6984***	1.6995***	1.8370***	1.7620***
Intercept	(0.0207)	(0.0420)	(0.0638)	(0.0760)	(0.0764)	(0.0804)	(0.0854)
c_daded					0.0019	0.0011	0.0016
c_daded					(0.0055)	(0.0054)	(0.0054)
c_dep		0.0067	0.0081	0.0071	0.0070	-0.0170	-0.0243
c_dep		(0.0167)	(0.0165)	(0.0155)	(0.0157)	(0.0161)	(0.0162)
c_edu	0.1019***	0.0957***	0.0895***	0.0700***	0.0703***	0.0663***	0.0678***
c_edu	(0.0082)	(0.0083)	(0.0083)	(0.0092)	(0.0093)	(0.0094)	(0.0163)
c_edu:female							0.0189
c_edu:female							(0.0149)
c_edu_sq						0.0030*	0.0052**
c_edu_sq						(0.0016)	(0.0021)
c_exper	0.0045**	0.0039**	0.0043**	0.0037**	0.0036**	0.0085***	0.0084***
c_exper	(0.0019)	(0.0019)	(0.0018)	(0.0018)	(0.0018)	(0.0022)	(0.0022)
c_exper:c_edu							0.0011*
c_exper:c_edu							(0.0006)
c_exper_sq						-0.0005***	-0.0005***
c_exper_sq						(0.0001)	(0.0001)
c_momed					0.0018	0.0020	0.0009
c_momed					(0.0063)	(0.0061)	(0.0061)
c_sib					0.0040	0.0053	0.0019
c_sib					(0.0156)	(0.0153)	(0.0153)
c_ten	0.0226***	0.0174***	0.0167***	0.0136***	0.0137***	0.0196***	0.0184***
c_ten	(0.0033)	(0.0032)	(0.0032)	(0.0030)	(0.0031)	(0.0050)	(0.0050)
c_ten_sq						-0.0004	-0.0004*
c_ten_sq						(0.0002)	(0.0002)
female		-0.2848***	-0.2944***	-0.2643***	-0.2645***	-0.2593***	-0.1366**
female		(0.0408)	(0.0401)	(0.0422)	(0.0423)	(0.0420)	(0.0604)
itai				-0.3533***	-0.3512***	-0.3345***	-0.3122***
itai				(0.0781)	(0.0787)	(0.0770)	(0.0772)
itarch				-0.3144***	-0.3131***	-0.2916***	-0.2909***
itarch				(0.0613)	(0.0620)	(0.0607)	(0.0606)
itclient				-0.0740	-0.0722	-0.0710	-0.0734
itclient				(0.0673)	(0.0677)	(0.0663)	(0.0662)
itprog				-0.0869	-0.0879	-0.0787	-0.0575

itprog				(0.0728)	(0.0731)	(0.0719)	(0.0718)
itqual				-0.1482	-0.1472	-0.1293	-0.1331
itqual				(0.1017)	(0.1021)	(0.1004)	(0.0999)
itrd				0.1761***	0.1751***	0.1563***	0.1507***
itrd				(0.0553)	(0.0557)	(0.0552)	(0.0550)
itreq				0.0449	0.0471	0.0543	0.0400
itreq				(0.0982)	(0.0992)	(0.0971)	(0.0968)
ittech				0.0232	0.0217	0.0317	0.0470
ittech				(0.0650)	(0.0654)	(0.0640)	(0.0638)
married		0.1063**	0.1240***	0.0843**	0.0829*	0.0371	0.1612***
married		(0.0452)	(0.0447)	(0.0427)	(0.0428)	(0.0430)	(0.0597)
married:female							-0.2139***
married:female							(0.0763)
metro			0.1675***	0.1604***	0.1591***	0.1530***	0.1369***
metro			(0.0453)	(0.0434)	(0.0435)	(0.0426)	(0.0430)
metro:c_edu							-0.0132
metro:c_edu							(0.0159)
nonprev		-0.0125	-0.0144	-0.0131	-0.0127	-0.0328	-0.0438
nonprev		(0.0654)	(0.0645)	(0.0615)	(0.0617)	(0.0607)	(0.0606)
norcen			-0.0268	-0.0350	-0.0348	-0.0372	-0.0431
norcen			(0.0556)	(0.0523)	(0.0525)	(0.0513)	(0.0510)
south			-0.0096	-0.0022	-0.0013	-0.0340	-0.0412
south			(0.0528)	(0.0500)	(0.0501)	(0.0495)	(0.0492)
west			0.1275**	0.1230**	0.1228**	0.1018*	0.0887
west			(0.0619)	(0.0585)	(0.0587)	(0.0577)	(0.0574)
wrkhm				-0.1029	-0.1035	-0.0793	-0.0845
wrkhm				(0.0649)	(0.0652)	(0.0643)	(0.0641)

Table 4. Fit statistics for the nested sequence.

Model	Parameters	R-squared	Adj. R-squared	AIC	BIC	F p-value
M1 Human capital	4	0.3057	0.3017	714.3654	731.4266	0.0
M2 + Demographics	8	0.3789	0.3706	663.7024	697.8248	0.0

M3 + Geography	12	0.4062	0.3935	648.0654	699.249	0.0
M4 + Job function	21	0.4864	0.466	589.8072	679.3786	0.0
M5 Full main effects	24	0.4868	0.4633	595.3802	697.7475	0.0
M6 + Quadratics	27	0.5127	0.4873	574.1	689.2631	0.0
M7 + Interactions	31	0.5243	0.4955	569.4312	701.6555	0.0

The human-capital block alone explains 31% of the variation in log salary. Adding the demographic, geographic and job-function controls raises the R-squared to 49%, and the quadratic and interaction terms add a further, statistically significant increment to an adjusted R-squared of 0.495. The joint tests in Table 5 confirm that every block contributes significantly, including the quadratic block (F = 9.05, p < 0.001) and the interactions (F = 3.02, p = 0.018).

Table 5. Joint significance of variable blocks in the preferred model.

Block	F statistic	p-value	Significant at 5%
Human capital (edu, exper, ten)	19.4026	0.0	True
Demographic controls	7.9868	0.0	True
Geographic controls	4.8026	0.0008	True
IT work-groups	7.9282	0.0	True
IT divisions	4.0128	0.0187	True
Quadratic human capital	9.0532	0.0	True
Interactions	3.0167	0.0178	True

5.3 Quadratic effects

The quadratic terms are the formal test of diminishing returns. The coefficient on experience-squared is negative and highly significant while the linear experience coefficient is positive, so the experience-earnings profile is concave and peaks within the sample. Evaluated at the mean of the other variables, the profile turns at approximately 26 years of experience: each extra year of experience is worth about 0.85% at the average experience level but nothing at the peak, and slightly negative beyond it. This is exactly the pattern Mincer's theory predicts and the pattern visible in Figure 5.

Tenure is also concave in principle, but the quadratic term is only weakly significant and the implied turning point (roughly 27 years of tenure) lies in the extreme upper tail of a variable whose median is only two years. We report the term for completeness but do not treat the concavity of tenure as an established finding. The education quadratic is positive and significant in the full sample; however, it is not robust. Dropping the handful of employees with implausibly few years of schooling (fewer than eight) reduces the coefficient to the margin of significance (p = 0.072), and trimming the top and bottom 1% of salaries does the same (p = 0.091). We therefore do not claim that returns to education accelerate; the safer reading is that the education effect is essentially linear over the range in which most of the data lie.

5.4 Interaction effects

Four interactions were tested. The interaction between gender and marital status is strong and materially important: the marriage premium is a male phenomenon. Married men earn about 17.5% more than comparable unmarried men, whereas for women the marriage effect is a statistically indistinguishable -5.1%. Put differently, the conditional gender gap is about 12.8% among unmarried employees but widens to roughly 29.6% among married employees, and the difference between those two gaps is significant (p = 0.012). Figure 9 illustrates the estimated profiles.

The remaining interactions are weaker. The interaction between education and experience is positive, which is consistent with education and experience being complements, but it does not quite reach conventional significance with robust standard errors (p = 0.107), and we report it as suggestive. The interaction between education and gender is small and insignificant, indicating that the returns to a year of schooling are similar for men and women in this sample; the gender gap therefore shows up as a shift in the intercept rather than as a difference in the slope of the education profile. The interaction between metro location and education is also insignificant.

Figure 9. Estimated interaction effects (other regressors at their sample means)

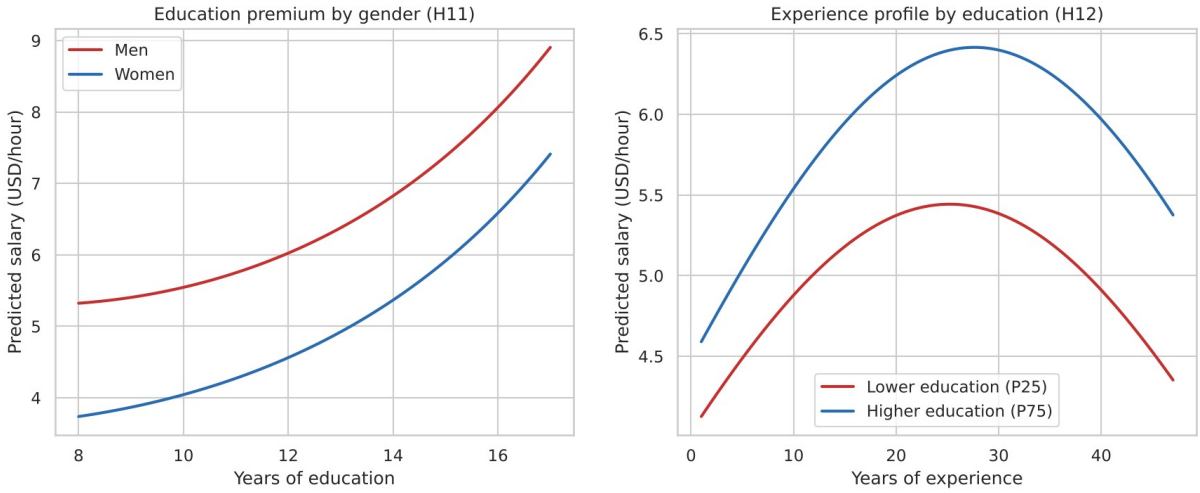


Figure 9. Estimated interaction effects, holding the other regressors at their sample means. Left: the education profile is higher for men but the slopes are statistically indistinguishable. Right: the experience profile is concave for both education groups, and the high-education profile lies above the low-education one throughout.

5.5 Control variables

The control variables are included because omitting them would attribute to the variables of interest variation that is in fact due to something else. The metropolitan indicator is a compensating-differential control: metro employees earn about 14.7% more (p = 0.004), which is consistent with a higher local cost of living rather than with superior productivity. The three province indicators (with the eastern province omitted) are jointly significant (F = 4.80, p = 0.001) but individually weak, indicating that the metro/non-metro distinction captures most of the geographic variation.

The job-function and division controls matter a great deal and are jointly highly significant (F = 7.93 for the work-groups, F = 4.01 for the divisions). R&D employees earn about 16.3% more than otherwise comparable employees (p = 0.008), consistent with the higher education concentration in that division seen in Figure 2. By contrast, IT architecture is associated with about 25.2% lower pay (p < 0.001) and AI-exposed work with about 26.8% lower pay (p < 0.001). The AI result is one of the most striking in the study and is discussed in the recommendations.

The family-background variables - mother's education, father's education and number of siblings - are individually and jointly insignificant. Their inclusion does not materially change the

education coefficient (0.0700 in M4 without them against 0.0703 in M5 with them), which is reassuring: the measured return to schooling is not simply a proxy for family background. Dependents and work-from-home status are likewise insignificant once the other controls are included.

5.6 Verification of model assumptions

Table 6 reports the diagnostic tests for the preferred model and for the models that precede it. Figure 8 shows the corresponding residual plots.

Table 6. Diagnostic tests (M7 unless stated).

Model	Test	Statistic	p-value	Conclusion
M4 + Job function	Jarque-Bera (residual normality)	691.3623763084581	7.457057413977067e-151	residuals non-normal
M4 + Job function	Breusch-Pagan (heteroskedasticity)	22.657026969253018	0.3059472991299097	homoskedastic
M4 + Job function	White / Wooldridge (heteroskedasticity)	1.2767241073517894	0.2798175394037146	homoskedastic
M4 + Job function	Ramsey RESET (functional form)	6.10680057899018	0.0137958067465583	misspecified
M4 + Job function	RESET, HC3-robust	7.900600008860964	0.0049418402129331	misspecified
M4 + Job function	Condition number of X	113.97376045826076		elevated, but scale-sensitive: see VIF table
M4 + Job function	Cook's D: share > 4/n (max D)	0.0513307984790874		max D = 0.094; no dominating observation
M7 + Interactions	Jarque-Bera (residual normality)	948.4816620879502	1.09603389352673 31e-206	residuals non-normal
M7 + Interactions	Breusch-Pagan (heteroskedasticity)	32.30258782199202	0.3535896557976583	homoskedastic
M7 + Interactions	White / Wooldridge (heteroskedasticity)	1.602586645928115	0.202362235886781	homoskedastic
M7 + Interactions	Ramsey RESET (functional form)	6.8576432102554294	0.0090974311423878	misspecified
M7 + Interactions	RESET, HC3-robust	8.201239911235817	0.0041861771993837	misspecified
M7 + Interactions	Condition number of X	2402.586704031782		elevated, but scale-sensitive: see VIF table
M7 + Interactions	Cook's D: share > 4/n (max D)	0.0627376425855513		max D = 0.096; no dominating observation

Figure 8. Assumption diagnostics for the preferred specification (M7)

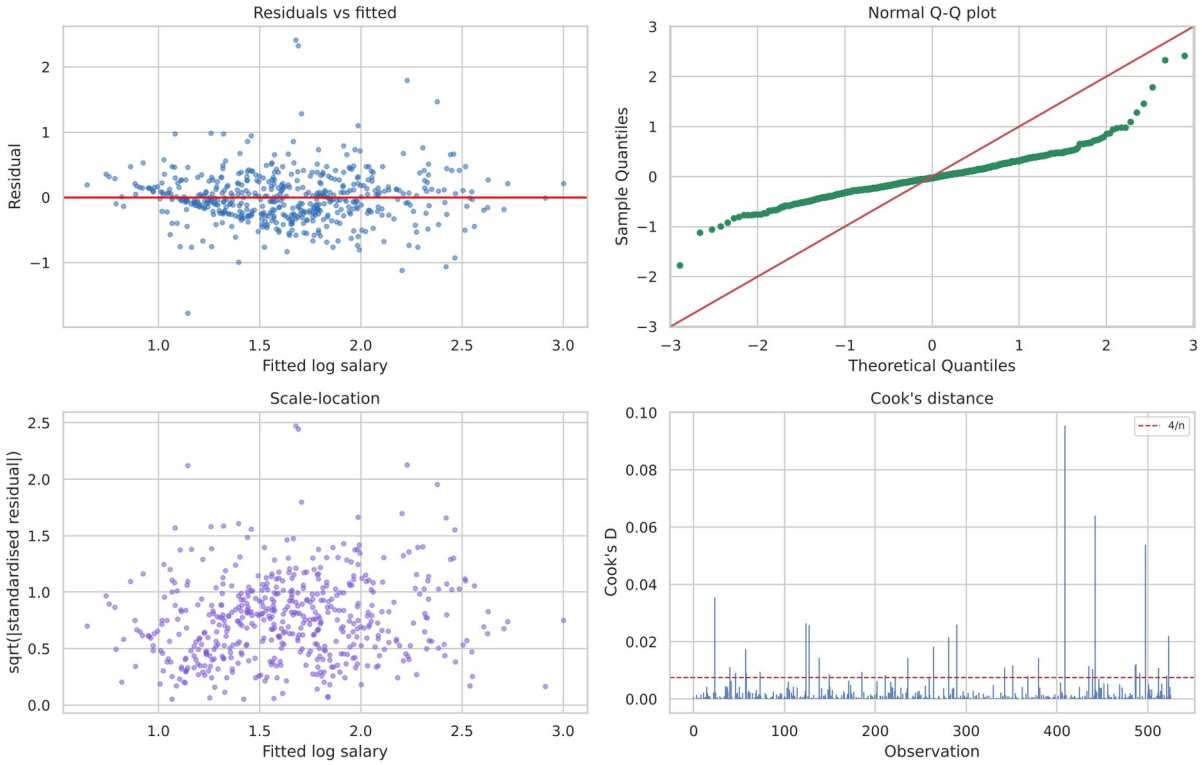


Figure 8. Residual diagnostics for the preferred specification. Residuals show no systematic pattern against fitted values; the Q-Q plot reveals fat tails rather than a shift; the scale-location plot shows mild but unsystematic spread; and the largest Cook's distance is well below one, so no single observation drives the estimates.

- Linearity and functional form. The Ramsey RESET test rejects the null of correct specification ($p = 0.009$, and $p = 0.004$ when computed with heteroskedasticity-robust covariance). This is the one classical assumption that is not satisfied, and it is investigated explicitly in Section 5.7.
- Normality of the errors. The Jarque-Bera test rejects normality decisively ($p < 0.001$). The Q-Q plot shows that the failure is one of fat tails rather than skew: a small number of employees earn salaries far above or far below prediction. Normality is not required for the OLS estimator to be unbiased or consistent, and with 526 observations the central limit theorem makes the usual tests approximately valid. We nevertheless report heteroskedasticity- and specification-robust standard errors so that inference does not depend on the normality assumption.
- Influence. The largest Cook's distance is 0.096, far below the conventional threshold of one. Although roughly 6% of observations exceed the 4/n flag, no single observation dominates. Dropping the ten most influential observations raises the adjusted R-squared from 0.495 to 0.579 and leaves every substantive conclusion intact.

5.7 Specification, multicollinearity and heteroskedasticity

Multicollinearity. Table 7 reports the variance-inflation factors for the preferred model. The largest is 6.0 for education, well below the conventional threshold of 10, and every other variable is below 5. The tolerance of education is 0.166, so 83% of the variation in education is independent of the other regressors. The textbook symptom of collinearity - a high R-squared alongside individually insignificant coefficients - does not appear here: the human-capital variables are all individually significant. We note that before mean-centring, education and its square had variance-inflation factors above 80; the reported figures are therefore the ones that matter, and they are acceptable.

Table 7. Variance-inflation factors, preferred model.

Variable	VIF	Tolerance	Assessment
c_edu	6.015	0.166	moderate (>5)
c_ten	4.214	0.237	ok
metro:c_edu	3.948	0.253	ok
married:female	3.524	0.284	ok
c_ten_sq	3.458	0.289	ok
female	2.93	0.341	ok
c_exper	2.742	0.365	ok
married	2.736	0.365	ok
itclient	2.708	0.369	ok
c_exper_sq	2.489	0.402	ok
itarch	2.422	0.413	ok
c_exper:c_edu	2.318	0.431	ok
itrd	2.266	0.441	ok
c_edu_sq	1.957	0.511	ok
ittech	1.826	0.548	ok
south	1.765	0.567	ok
itai	1.738	0.575	ok
itprog	1.681	0.595	ok
c_edu:female	1.675	0.597	ok
wrkhm	1.599	0.625	ok
norcen	1.569	0.637	ok
west	1.48	0.676	ok
itqual	1.345	0.743	ok
c_dep	1.342	0.745	ok
itreq	1.315	0.761	ok
c_momed	1.228	0.815	ok
c_daded	1.225	0.816	ok
metro	1.197	0.836	ok
nonprev	1.092	0.916	ok
c_sib	1.04	0.962	ok

Heteroskedasticity. The Breusch-Pagan test does not reject homoskedasticity (p = 0.354), and neither does the White/Wooldridge test (p = 0.202). The scale-location plot in Figure 8 shows no funnel. We therefore do not need to correct for heteroskedasticity; the HC3 results reported in Table 10 show that doing so anyway changes the standard errors only trivially, which is reassuring.

Specification. The RESET rejection is the one genuine problem, so we investigated it rather than assuming it away. Three pieces of evidence suggest that the misspecification is mild and diffuse

rather than the omission of a specific important variable. First, a partial-residual analysis tested each continuous regressor for unmodelled curvature by testing whether its square and cube add explanatory power to the partial residuals; no variable showed evidence of non-linearity (smallest p-value 0.058 for experience; see Table 8). Second, the rejection survives dropping the most influential observations and is not an artefact of outliers. Third, and most tellingly, a deliberately flexible machine-learning benchmark does not beat the linear model out-of-sample: five-fold cross-validated R-squared is 0.416 for a linear model in the raw variables, 0.448 for the preferred quadratic-and-interaction design, and only 0.361 for gradient boosting. If an important non-linearity or interaction had been omitted, a flexible learner should have found it and improved on the linear model; it does not. We therefore read the RESET rejection as a symptom of the fat-tailed error distribution rather than of substantive misspecification, and we address it by reporting heteroskedasticity-robust standard errors and by demonstrating robustness across the alternative specifications in Table 10.

Table 8. Partial-residual tests for unmodelled non-linearity (joint F-test on the square and cube of each regressor).

Regressor	F statistic	p-value	Evidence of non-linearity
c_edu	0.4407	0.6438	False
c_exper	2.8621	0.0580	False
c_ten	0.7735	0.4619	False
c_dep	0.1383	0.8708	False
c_momed	0.3744	0.6879	False
c_daded	0.3730	0.6889	False
c_sib	1.8559	0.1573	False

Table 9. Cross-validated performance of a flexible benchmark (five-fold CV R-squared on log salary).

Model	Cross-validated R-squared
OLS, linear in X	0.4157
OLS, quadratics + interactions (M7 design)	0.4477
Gradient boosting (flexible benchmark, raw X)	0.3608

Table 10. Alternative specifications.

Model	N	R-squared	Adj. R-squared	AIC	BIC
M1 Human capital	526	0.3057	0.3017	714.3654	731.4266
M2 + Demographics	526	0.3789	0.3706	663.7024	697.8248
M3 + Geography	526	0.4062	0.3935	648.0654	699.249
M4 + Job function	526	0.4864	0.466	589.8072	679.3786
M5 Full main effects	526	0.4868	0.4633	595.3802	697.7475
M6 + Quadratics	526	0.5127	0.4873	574.1	689.2631
M7 + Interactions	526	0.5243	0.4955	569.4312	701.6555
M8 M7 with HC3 robust SE	526	0.5243	0.4955	569.4312	701.6555
M9 DV in levels	526	0.2905	0.2475	3190.3484	3322.5728
M10 Complete-case sample	482	0.5284	0.497	535.6447	665.161

M11 1% trimmed salary	514	0.5694	0.5426	383.0231	514.5321

Table 10 shows that the results are not an artefact of either the functional form or the imputation. Estimating the same specification with salary in levels instead of logarithms reduces the adjusted R-squared from 0.495 to 0.248 without changing any sign; restricting the sample to the 482 complete cases gives an adjusted R-squared of 0.497 against 0.495; and trimming the extreme 1% of salaries raises it to 0.543, indicating that outliers add noise rather than signal. The preferred model with HC3 robust standard errors (M8) has identical point estimates, and the significance of the key variables - education, experience, tenure, the gender gap, the marriage premium and the AI penalty - is unchanged.

5.8 Economic magnitudes

Table 11 restates the preferred model's coefficients in the form the client can act on. Because the dependent variable is in logarithms, a coefficient b corresponds to a proportional effect of 100 x (exp(b) - 1) percent.

Table 11. Estimated economic magnitudes (preferred model, HC3 p-values).

Effect	Log points	Approx. % effect	p-value
Women, unmarried (vs comparable men)	-0.1366	-12.7668	0.0345
Women, married (vs comparable married men)	-0.3504	-29.5625	0.0
Married men (vs comparable unmarried men)	0.1612	17.4926	0.0114
Married women (vs comparable unmarried women)	-0.0527	-5.1292	0.3588
One extra year of education (at mean)	0.0678	7.0188	0.0004
One extra year of experience (at mean)	0.0084	0.8483	0.0001
One extra year of tenure (at mean)	0.0184	1.8601	0.0003
Metro location	0.1369	14.6762	0.0037
IT architecture (vs other work-groups)	-0.2909	-25.2439	0.0
IT R&D division	0.1507	16.2642	0.0075
AI-exposed IT work	-0.3122	-26.819	0.0001

5.9 Summary of hypothesis tests

Table 12. Verdicts on the stated hypotheses.

H#	Hypothesis	Term	Estimate	p-value	Verdict
H1	Education raises pay	c_edu	0.0678	0.0004	Supported
H2	Experience raises pay but with diminishing returns	c_exper	0.0084	0.0001	Supported
H2b	The experience-pay profile is	c_exper_sq	-0.0005	0.0003	Supported

	concave				
H3	Firm-specific tenure raises pay	c_ten	0.0184	0.0003	Supported
H4	Women earn less than comparable men	female	-0.1366	0.0345	Supported
H5	Married employees earn a premium	married	0.1612	0.0114	Supported
H6	Metro employees earn a premium	metro	0.1369	0.0037	Supported
H7	Pay differs across IT work-groups and divisions	itreq = 0, itprog = 0, itqual = 0, itarch = 0		0.0	Supported
H7b	R&D and tech-support divisions shift pay	itrd = 0, ittech = 0		0.026	Supported
H8	Non-privileged employees earn less	nonprev	-0.0438	0.4851	Not supported
H9	Working from home is associated with lower pay	wrkhm	-0.0845	0.221	Not supported
H9b	AI-exposed IT work is associated with lower pay	itai	-0.3122	0.0001	Supported
H10	Parental education and siblings matter	c_momed	0.0009	0.9001	Not supported
H11	The education premium differs by gender	c_edu:female	0.0189	0.2385	Not supported
H12	Education and experience are complements	c_exper:c_edu	0.0011	0.1071	Not supported
H13	The marriage premium differs by gender	married:female	-0.2139	0.0116	Supported

Eleven of the sixteen hypotheses are supported and five are not. The human-capital core of the model - education, experience (concave), and tenure - is confirmed, as are the metro premium, the R&D premium and the AI penalty. The five hypotheses that fail are informative in their own right: non-privileged status and work-from-home carry no independent pay effect once job function, location and human capital are controlled; family background adds nothing once the employee's own education is measured (the three family variables are also jointly insignificant, $F = 0.06$, $p = 0.98$); and the education-gender and education-experience interactions are not significant, the latter being only suggestive ($p = 0.107$).

6. Recommendations

The estimates support the following recommendations to HCL Technologies.

- Investigate the married-women pay gap as a matter of priority. It is the largest and most precisely estimated differential in the study: married women earn an estimated 29.6% less

than comparable married men, while unmarried women earn 12.8% less than comparable unmarried men. The gap is a shift in the pay level rather than a difference in returns to education or experience, so it cannot be explained by women accumulating less human capital. A structured audit comparing like-for-like roles, grades and performance ratings within marital-status groups is the appropriate first step.

- Review the compensation of IT architecture and AI-exposed roles. These two groups carry the largest negative coefficients in the model (about -25% and -27% respectively). If these reflect obsolete skills this is a retention risk in the short run and a reskilling obligation in the medium run; if they reflect the age composition of those groups, the model should be extended with age, which the current extract does not contain.
- Preserve the R&D premium and treat it as a retention instrument. R&D employees earn an estimated premium of about 16%. Given the 0.50 correlation between R&D membership and education, part of this premium is a return to screening; the premium should be monitored against market benchmarks rather than treated as a permanent entitlement.
- Expect the experienced-workforce premium to flatten. The experience profile peaks at roughly 26 years. Pay policy that assumes seniority continues to command increases beyond that point is not supported by the data; progression beyond the peak should be justified by role and performance rather than by service length alone.
- Do not treat work-from-home or non-privileged status as pay determinants. Both are statistically indistinguishable from zero once job function, location and human capital are controlled. Any policy that prices remote work differently would therefore be justified by evidence outside this dataset rather than by the salary structure it reveals.
- Improve the data-collection process. Twenty-nine cells in the extract contradicted the data dictionary, including salary values of 1,111 and education values of 1,001. Range and type validation at the point of entry would remove the need for the imputation documented in Section 2 and would strengthen the reliability of any subsequent audit.

These recommendations follow directly from the estimated coefficients and the diagnostics. Where the evidence is weaker - the education interaction, the tenure curvature - we have said so rather than overstating the case.

7. References

Becker, G. S. (1964). Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education. New York: Columbia University Press.

Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). Regression Diagnostics: Identifying Influential Data and Sources of Collinearity. New York: Wiley.

Breusch, T. S., & Pagan, A. R. (1979). A simple test for heteroscedasticity and random coefficient variation. Econometrica, 47(5), 1287-1294.

Cook, R. D. (1977). Detection of influential observation in linear regression. Technometrics, 19(1), 15-18.

Halvorsen, R., & Palmquist, R. (1980). The interpretation of dummy variables in semilogarithmic equations. American Economic Review, 70(3), 474-475.

Jarque, C. M., & Bera, A. K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. Economics Letters, 6(3), 255-259.

Kennedy, P. E. (1981). Estimation with correctly interpreted dummy variables in semilogarithmic equations. American Economic Review, 71(4), 801.

Mincer, J. (1974). Schooling, Experience, and Earnings. New York: Columbia University Press for NBER.

Ramsey, J. B. (1969). Tests for specification errors in classical linear least-squares regression analysis. Journal of the Royal Statistical Society: Series B, 31(2), 350-371.

Thursby, J. G., & Schmidt, P. (1977). Some properties of tests for specification error in a linear regression model. Journal of the American Statistical Association, 72(359), 635-641.

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroscedasticity. Econometrica, 48(4), 817-838.

Wooldridge, J. M. (2020). Introductory Econometrics: A Modern Approach (7th ed.). Boston: Cengage Learning.

Data source: ITEmp.xlsx (HCL Technologies employee extract, supplied for this assignment). All computations were performed in Python (pandas, statsmodels, scikit-learn); the analysis scripts (code/run_all.py) accompany this report.